

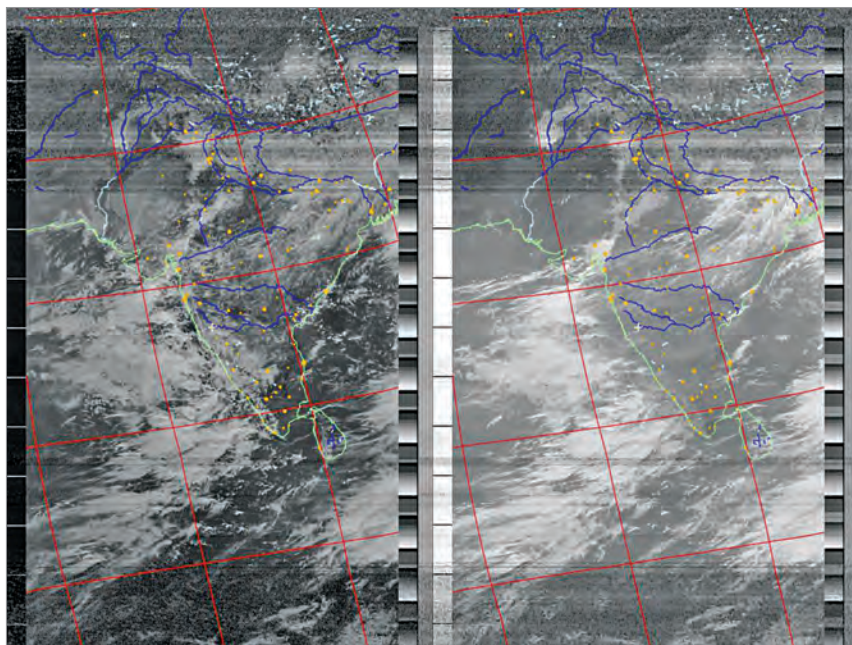


Fig. 25: A RAW image successfully received from the satellite showing two channels with graphics overlay after processing

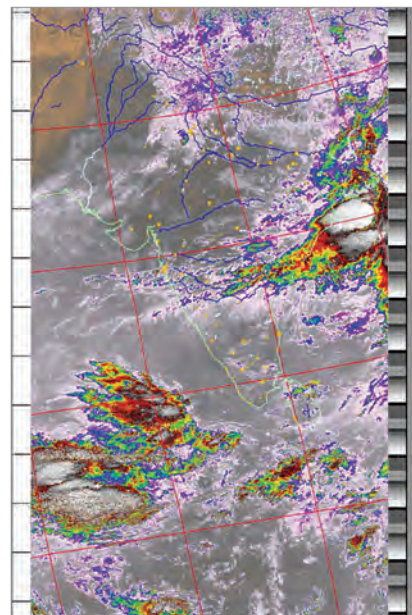


Fig. 26: One of the processed output images is the colour-coded cloud temperatures

respectively. You can use any one of these. The author uses GPredict as shown in Fig. 22.

Alternately, the WXtoImg software also produces a satellite pass list from the main File menu. You can use any of the above two tools to complement each other. It is suggested to save the satellite passes into a text file for future reference.

Analogue APT image reception

Now, you are ready to receive the actual satellite signals. Start your receiver at least five minutes before the arrival of the satellite on the horizon and tune to the satellite frequency. Keep the receiver bandwidth to a minimum of 50kHz. Wideband FM settings of 150kHz or more would simply add more noise to the signal.

Open the prediction software to look for satellite and its position on the map. With practice you should be able to learn well when your receiver starts picking up the signal.

Open the WXtoImg software and under the main File menu choose the submenu Record and then Auto Record option as shown in Fig. 23.

If the satellite is yet to arrive on

the horizon in your sky, the recorder in decoding software will wait and display the status accordingly. Once, the satellite appears, the software starts decoding the image from the received audio. You can watch the raw image as displayed by this software during the transit of a satellite as shown in Fig. 15.

Once the satellite pass is completed, the software automatically processes the raw image and then displays and stores the processed images. To display the raw images, the map overlays, and other features on the processed data, go to the main menu Options and choose Map Overlay Options submenu and follow the steps in the dialogue box as shown in Fig. 24.

A couple of processed images, after the signal has been received successfully from a satellite transit, are shown in Fig. 25 and Fig. 26.

There might be many features disabled in the freeware version, but you can experiment and try various image enhancement options. But the images obtained are beyond doubt sufficient for an experimenter to see the clouds and carry out basic sci-

tific studies. The author is thankful to the original authors/creators for their excellent work in making the freeware software available for the experimenters. The copyrights, where applicable, are honoured with thanks.

With the above steps-by-step information, the author is confident that this article would help and guide the readers in setting up their own weather satellite receiving station. Kindly note, this experimentation requires great patience and consistency in efforts to establish and maintain your weather satellite receiving station. Once you start receiving the images on a regular basis, you can identify the areas to improve and work towards achieving better and better results. **EFY**

Prasanna Waichal holds a master's degree in electronics and has more than 25 years of work experience in industrial R&D for design and development of specialised and strategic electronics systems and academic teaching and research at post-graduate levels, both in India and abroad. His passions include radio reception and studying solar activities and their effect on ionosphere on earth. He is also a licensed amateur radio operator with call-sign VU3OOL. He claims to have discovered a method to predict earthquakes using very-low-frequency (VLF) multi-path-multi-frequency propagation technique

